

Vertical APIs for Agentic AI

Routing the Right Context to the Right Expert

David vonThenen

AI/ML Engineer

     [@davidvonthenen](https://twitter.com/davidvonthenen)



David vonThenen

- Agentic Solutions
- Book In Progress
- AI Harness Wizard
- AI/ML Security Hacker
- Part-time GPU Owner

    [@davidvonthenen](https://twitter.com/davidvonthenen)



Agenda

- The Context Crisis
- Vertical APIs and MoEs
- Agentic AI & AI Governance
- **Live Demo: Importance of Context Routing!**
- Q & A

The Context Crisis

The Generalist Delusion...

Couple of Leading Cause of Failed AI Solutions:

- Mix Unrelated Authority Domains
- Use Stale/Incomplete Data in RAG

Confidently Incorrect Answers!

Example:

How is <Company XYZ> performing?

- Speculative Reddit Rumors
- Outdated SEC Earnings
- Stale News Articles

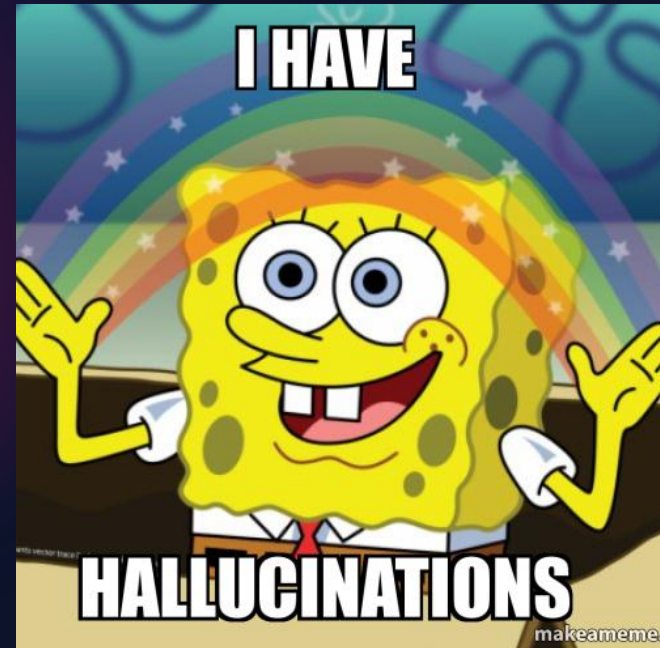


Image Attribution
Make-a-meme

<https://makeameme.org/meme/i-have-hallucinations>

Context Failure by Example

How is SpaceX Doing Financially & Reaching Its Goals?

Starlink is the Profit Engine

- Revenue: \$11.4B (+48% YoY, 39% Margin)

The Bull Case:

- Commercial Spaceflight Monopoly
- Global Connectivity Infrastructure

The Bear Case:

- \$1.77T Valuation
- "Priced for Absolute Perfection" = ???

Image Attribution

Whoopsie! A SpaceX Rocket Accidentally Crashed Into the Moon.

<https://www.popularmechanics.com/space/moon-mars/a73358498/moon-debris-falcon-9/>



The Real Crisis

Failure Mode Aren't Always Hallucinations

Often it is the:

- Wrong Corpus
- Wrong Workflow
- Wrong Authority (Fail)
- Wrong Policy Boundary (Fail)
- Wrong Evidence Standard (Fail)

The Model May Sound Right, But...
Has Wrong "World View" (No Perspective)

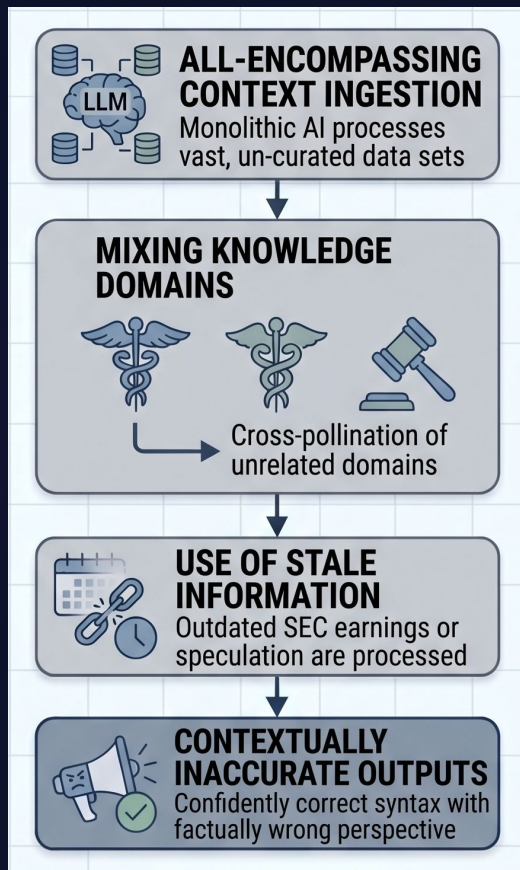


Image Attribution
Gemini Generated

Monolithic Apps, Monolithic LLMs

We Broke Monoliths Into Services Because...

- Boundaries Mattered
- Separation of Concerns
- Abstraction Is Helpful

We Are Relearning Those Lessons!

Old Problem:

One App Owned Everything

New Problem:

One Agent Handles Everything

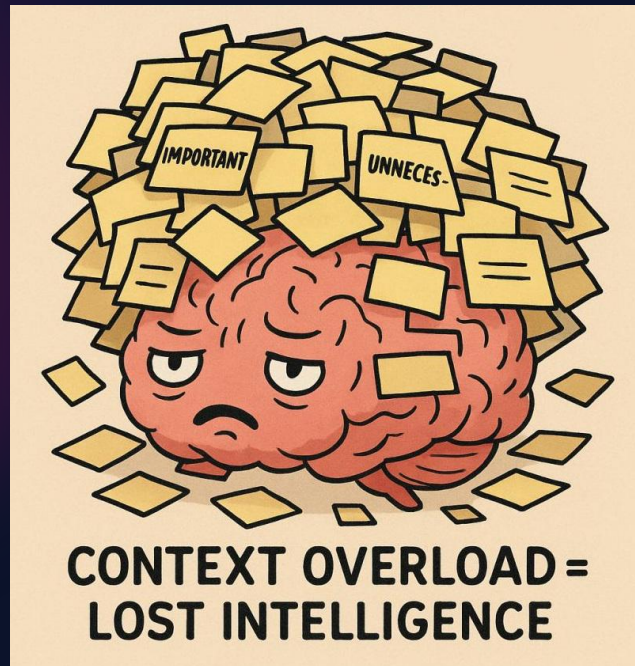


Image Attribution
kannadasan saminathan

https://www.linkedin.com/posts/kannadasan-saminathan_llm-context-overload-the-hidden-bottleneck-share-7370710632670711809-jAb4/

What are Mixture of Experts?

MoE layer



Image Attribution

<https://www.gate.com/learn/articles/what-a-decentralized-mixture-of-experts-mo-e-is-and-how-it-works/5073>

<https://medium.com/@sthanikamsanthosh1994/the-mixer-of-experts-moe-modern-architecture-for-divide-and-concur-learning-dbe10ffa8436>

Claude Code: Mostly Harness Code



Search Submit Donate Log in

Computer Science > Software Engineering

[Submitted on 14 Apr 2026 (v1), last revised 2 Jul 2026 (this version, v2)]

Dive into Claude Code: The Design Space of Today's and Future AI Agent Systems

[Jiacheng Liu](#), [Xiaohan Zhao](#), [Xinyi Shang](#), [Zhiqiang Shen](#)

Claude Code is an agentic coding tool that can run shell commands, edit files, and call external services on behalf of the user. This study describes its architecture by analyzing the publicly available source code and comparing it with two independent open-source AI agent systems, OpenClaw and Hermes Agent, that answer many of similar or even the same design questions. Our analysis identifies five human values, philosophies, and needs that motivate the architecture: human decision authority, safety, security, and privacy, reliable execution, capability amplification, and contextual adaptability. We then trace them through thirteen design principles to implementation choices. The core of the system is a simple while-loop that calls the model, runs tools, and repeats. Most of the code, however, lives in the systems around this loop: a permission system with seven modes and an ML-based classifier, a five-layer compaction pipeline for context management, four extensibility mechanisms (MCP, plugins, skills, and hooks), a subagent delegation and orchestration mechanism, and append-oriented session storage. Comparisons with OpenClaw and Hermes Agent show that the same design questions produce different answers across three deployment contexts. Claude Code emphasizes per-action safety, OpenClaw emphasizes perimeter-level access, and Hermes renders per-action approvals across many surfaces. At the runtime layer, Claude Code uses a single CLI loop. OpenClaw embeds the runtime within a gateway control plane, and Hermes

Access Paper:

[View PDF](#)
[HTML \(experimental\)](#)
[TeX Source](#)

 [view license](#)

Current browse context:

cs.SE

[< prev](#) | [next >](#)

[new](#) | [recent](#) | [2026-04](#)

Change to browse by:

cs

[cs.AI](#)

[cs.CL](#)

[cs.LG](#)

References & Citations

[NASA ADS](#)
[Google Scholar](#)
[Semantic Scholar](#)

Export BibTeX Citation

Bookmark



Image Attribution

Dive into Claude Code: The Design Space of Today's and Future AI Agent Systems

<https://arxiv.org/abs/2604.14228>

Small Language Models for the Win

Small Language Models are the Future of Agentic AI

Peter Belcak¹ Greg Heinrich¹ Shizhe Diao¹ Yonggan Fu¹ Xin Dong¹
Saurav Muralidharan¹ Yingyan Celine Lin^{1,2} Pavlo Molchanov¹
¹NVIDIA Research ²Georgia Institute of Technology
agents-research@nvidia.com

Abstract

Large language models (LLMs) are often praised for exhibiting near-human performance on a wide range of tasks and valued for their ability to hold a general conversation. The rise of agentic AI systems is, however, ushering in a mass of applications in which language models perform a small number of specialized tasks repetitively and with little variation.

Image Attribution

Small Language Models are the Future of Agentic AI

<https://arxiv.org/pdf/2506.02153>

Vertical APIs and MoEs

Vertical APIs = Domain + Authority

API Designed Specifically to Serve:

- A single, Niche Industry, or
- Specialized Business Function

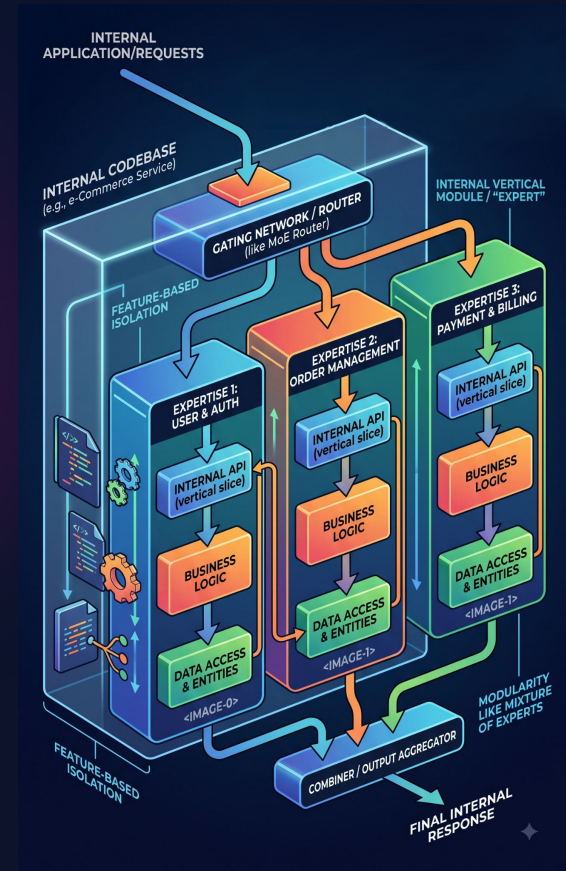
More Importantly, Vertical API Exposes:

- Single Domain, and
- Single Authority

Examples:

- Legal Clause Review
- Clinical Safety Triage
- Financial Risk Explanation

Image Attribution
Gemini Generated



MoEs Are Vertical APIs

Each Expert Domain Has Its Own:

- Corpus and Retrieval Strategy
- Access Policy
- Prompt Contract
- Evaluation/Validation Rules
- Audit Trail

Ex: Agents w/ Own Governance:

- Healthcare Expert
- Financials Expert

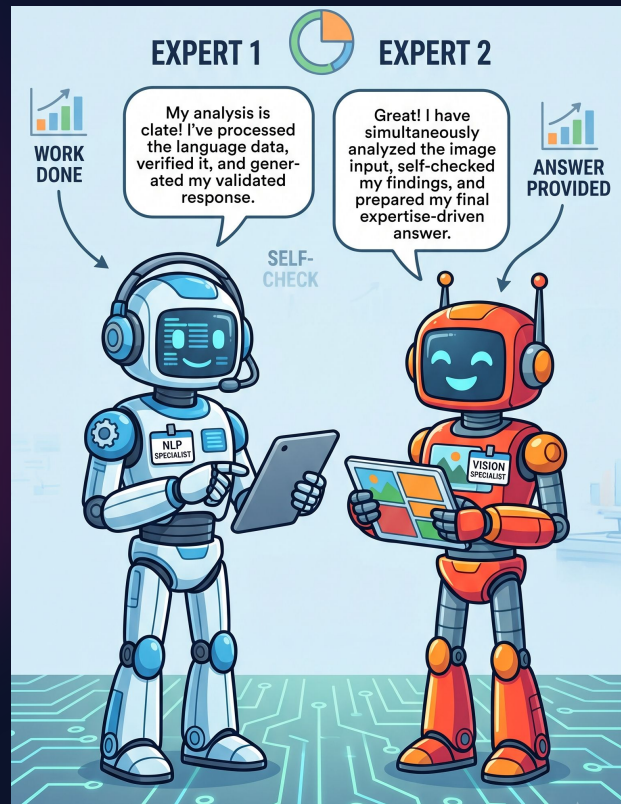


Image Attribution
Gemini Generated

Silos Aren't Bad... Yes, I Said It

In Heavily Regulated AI, Silos Enforce:

- Access Control
- Retention Policy
- Auditability
- Domain-Specific Validation

Silos Are GOOD In the **Right Context!**

- Silos Are Policy Attachment Points!
- AI Governance!
- People or Org Silos Are **BAD!**

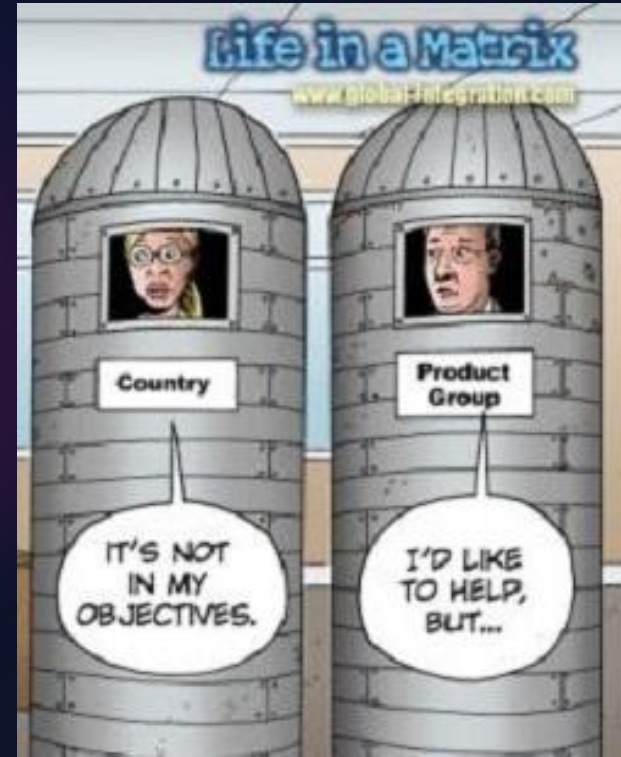


Image Attribution

Machiavelli predicted business silos in 1513, but that doesn't mean you have to be resigned to them

<https://thecontextofthings.com/2014/04/06/machiavelli-predicted-business-silos-in-1513-but-that-doesnt-mean-you-have-to-be-resigned-to-them/>

Vertical APIs for AI

The Endpoints:

- OpenAI-compatible API
- Agent2Agent
- MCP (Live Tools)

The Endpoint (aka MoE) Owns:

- Private Corpus and Metadata Filters
- Domain-Specific Prompts
- Governance Policy
- Citation Policy
- Response Validator



Image Attribution
Gemini Generated

Agentic AI and AI Governance

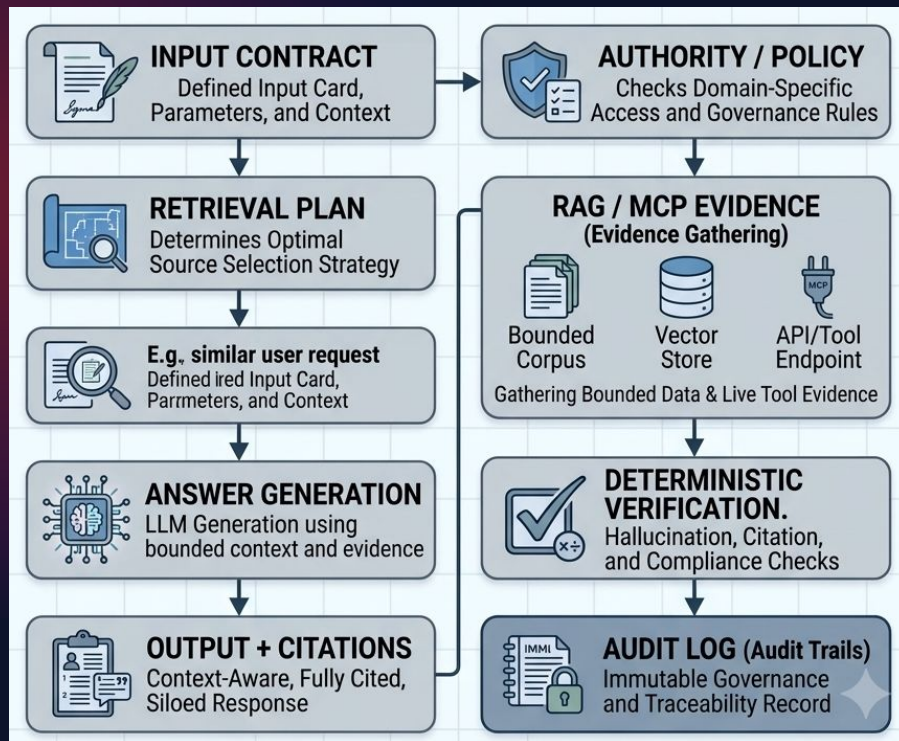
How Agents Become MoEs...

How to Transform Agents Into an Expert? Minimally...

- Input Contract
- Domain & Corpus
- Retrieval Layer
- Reasoning Policy
- Output Schema
- Evidence Requirements
- Audit & Logging

An EXPERT in its Field!

Image Attribution
Gemini Generated



The Orchestrator is the Traffic Cop

The Orchestrator is a Control Plane!

- Not a Knowledge Base
- Doesn't Need to Know Everything
- Doesn't "Contribute" to the Answer

It Needs to Know:

- What the User is Asking
- Which MoE Owns the Answer
- Which Policies Apply
- Determines If Answer Passes Review



Image Attribution

Princeton University Researcher - [Tool helps clear biases from computer vision](#)

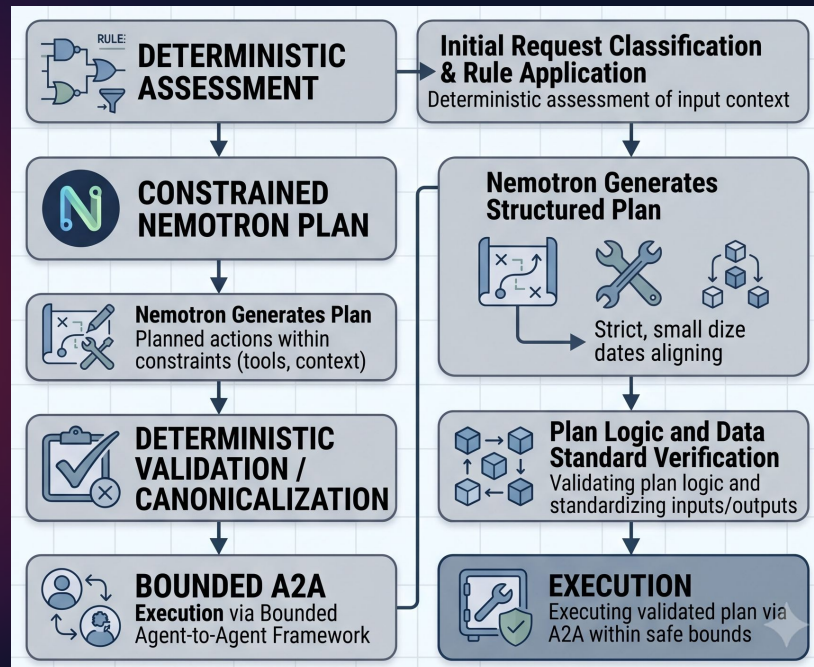
Molly Sharlach on October 1, 2020

Routing Strategies

Mechanisms for Routing:

- Deterministic Rules
- Fine-Tuned AI Model
- ML Classifiers
- Embedding Similarity
- Constrained LLM Router
- Human in the Loop

Truth Be Told, Boring Arch. Wins
Being **Deterministic** Will Go Far!



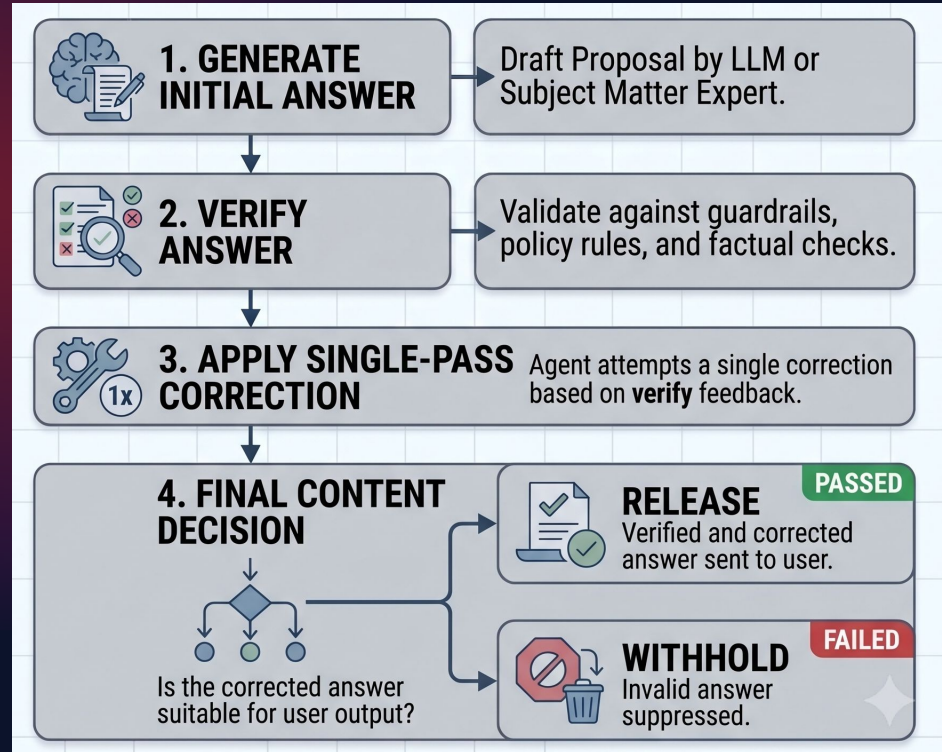
The "Self-Improvement" Loop

AKA the Bounded Correction Loop

The verifier checks:

- "Guardrails"
- Schema Validation
- Evidence Coverage
- Policy Compliance
- Citation Quality
- Domain Fit
- Unsafe Language

Image Attribution
Gemini Generated



The Verifier is NOT Optional

You Want to Have Checks for:

- Unsupported Claims
- Evidence Availability
- Missing Citations
- Invented Citations
- Cross-Domain Leakage
- Retry w/ Stronger Constraints
- Escalate Uncertainty

Deterministic Verifier = Authoritative



Image Attribution

Princeton University Researcher - [Tool helps clear biases from computer vision](#)

Molly Sharlach on October 1, 2020

Why MoEs Beat LLMs w/ Context?

IT'S SIMPLE: **CHAIN OF CUSTODY**

For Each API Call, We Observe:

- User Inputs
- Source Granularity
- Corpus-Specific Retrieval
- Tool Call Inputs/Outputs
- Policy Enforcement
- Evidence Traceability
- Agent Responses

HARNESS CODE!

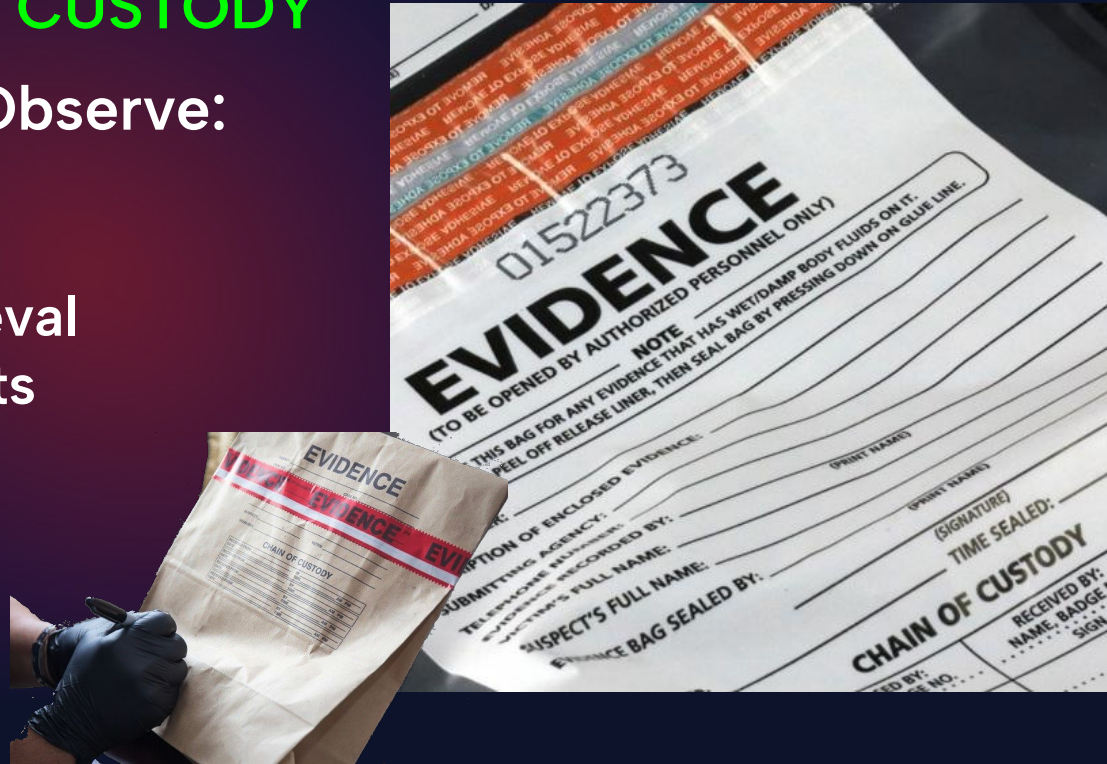


Image Attribution

What Is the Chain of Custody of Evidence?

<https://www.craftlegal.com/2021/01/20/what-is-the-chain-of-custody-of-evidence/>

Live Demo: A Concrete Example

Demo: Agentic RAG for the AI Field

"Theoretical" AI Company Evaluation Tool

- Orchestration Agent ([Nemotron Orchestrator](#))
 - AI Governance (Auditing, Observability, etc)
 - Context-based Routing
- News Agent
 - [Instaclustr OpenSearch](#): News Sources
 - MCP: [Tavily](#) (Agent Search API)
- Financials Agent
 - [Instaclustr OpenSearch](#): SEC Quarterly Filings
 - MCP: [FinnHub](#) API (Real-Time Quotes)

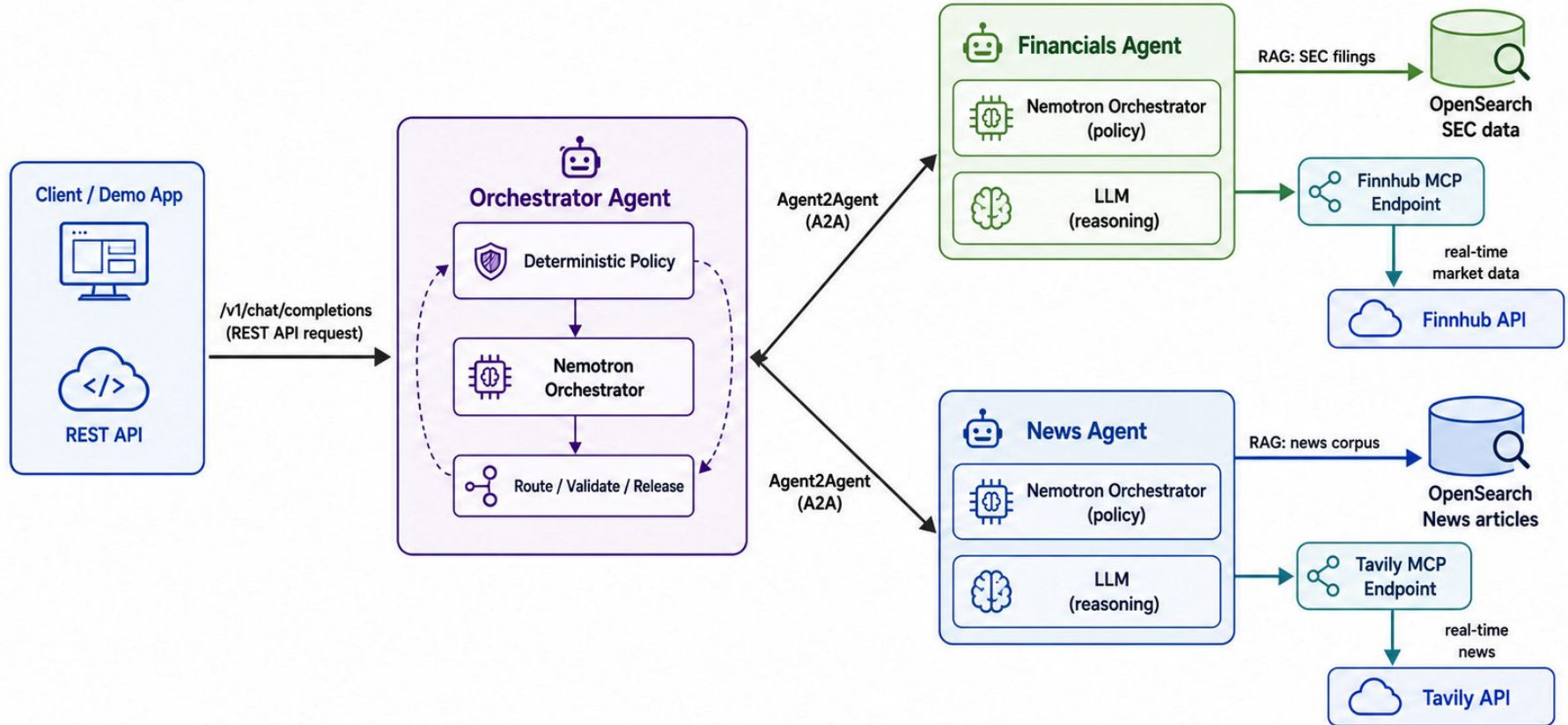
Image Attribution

Princeton University Researcher - [Tool helps clear biases from computer vision](#)

Molly Sharlach on October 1, 2020

Architecture Visualization

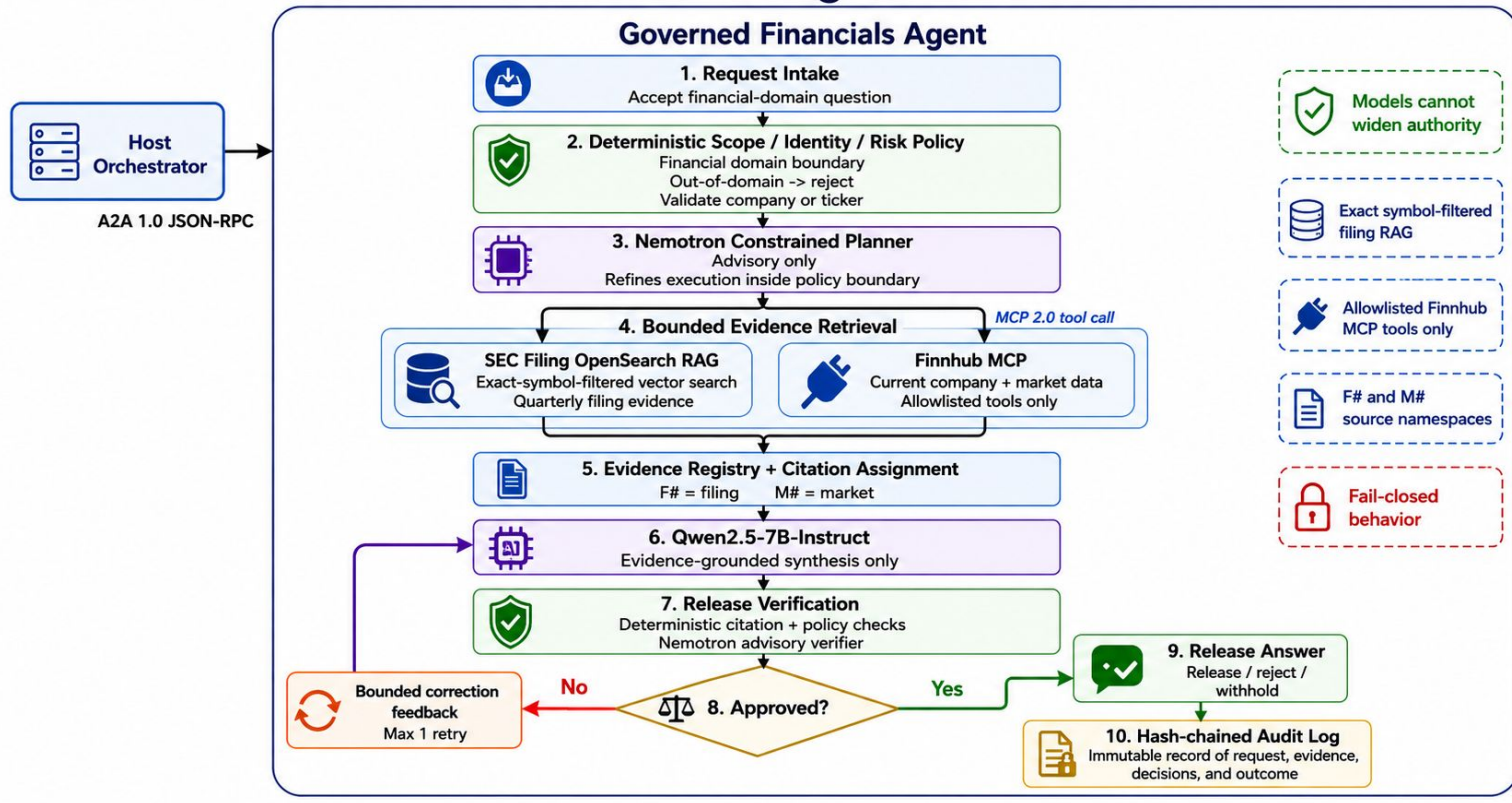
Image Attribution
Gemini Generated



Financials Agent Architecture

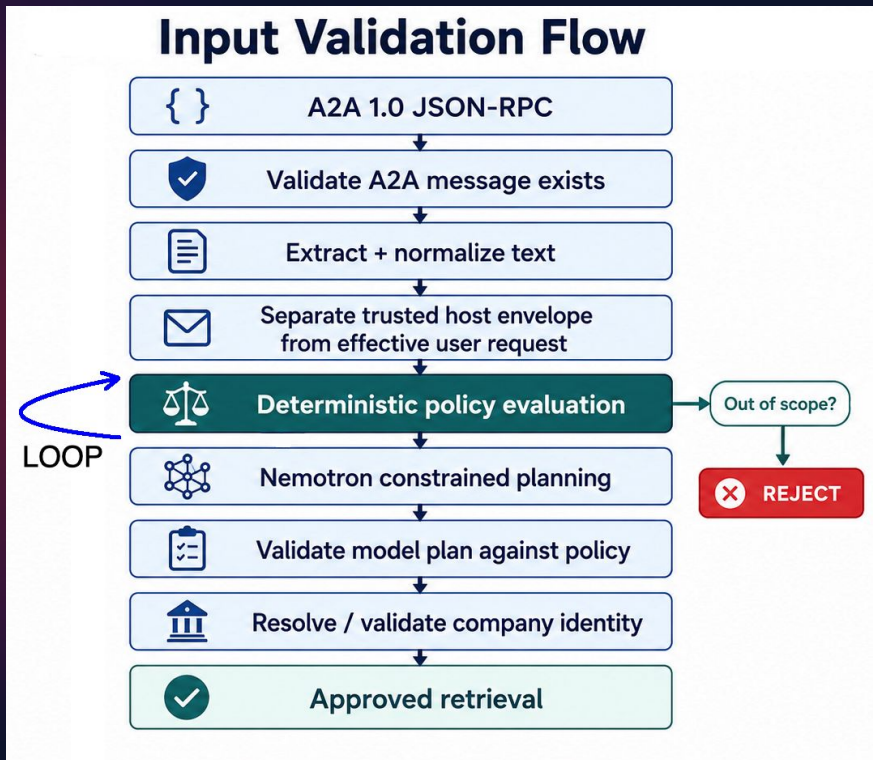
Image Attribution
Gemini Generated

Governed Financials Agent Architecture



To Be Agentic...

- **Input:**
 - Reject Requests Outside The Vertical
- **Planner:**
 - Uses Nemotron Orch.
 - Deterministic Plan
 - Identity is a Boundary
 - RAG and MCP in //
- **What is Done?**
 - Run All the Tools Authorized
 - Stop Loop = No New Evidence!



Data Accountability

Financial Agent

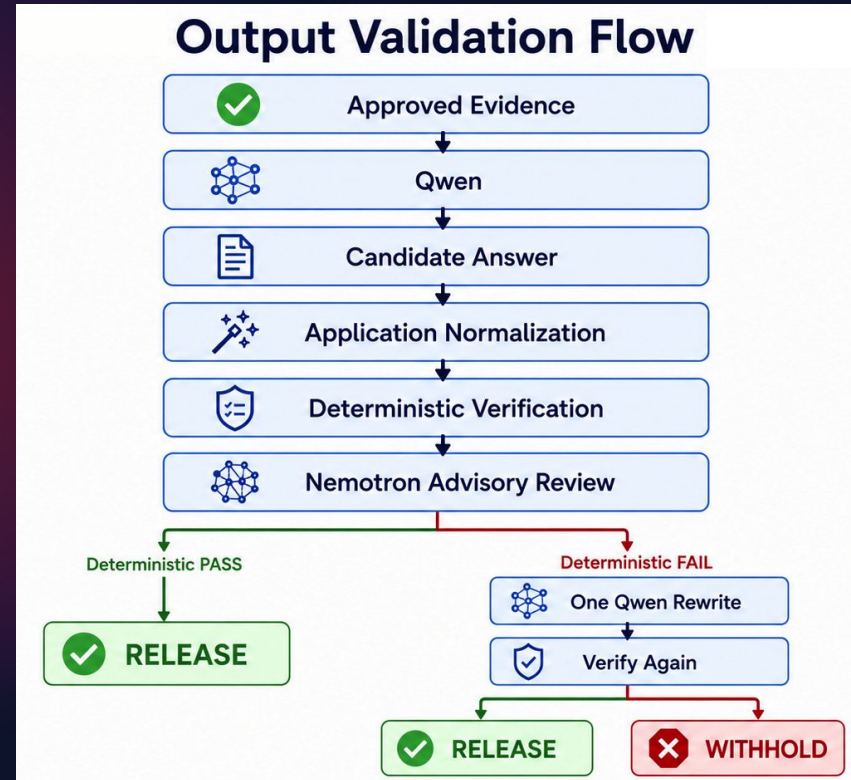
- |
- +--> [F1] SEC filing Q1
- +--> [F2] SEC filing Q2
- +--> [F3] SEC filing Q3
- +--> [M1] Finnhub Market Data 1
- +--> [M2] Finnhub Market Data 2
- +--> [M3] Finnhub Market Data 3

All Data (Embeddings or MCP)
Provides Citation Markers as
Evidence!

```
(2026-api-world) vonthd@vonthds-  
python -m src.query -q "What is NVDA stock price?"  
2026-08-26T19:22:21-0700 | INFO  
1:9002  
2026-08-26T19:22:42-0700 | INFO  
task_id=ce5308fb-0f4d-4138-a6ff-0  
## Financial assessment  
The current stock price of NVDA :  
  
## Evidence limits  
None identified.  
  
## Sources  
- [M1] Finnhub via MCP: get_stock_price  
https://finnhub.io/  
  
Audit ID: 576eb884-60b0-424d-88e
```

Validate Output **THEN** Release...

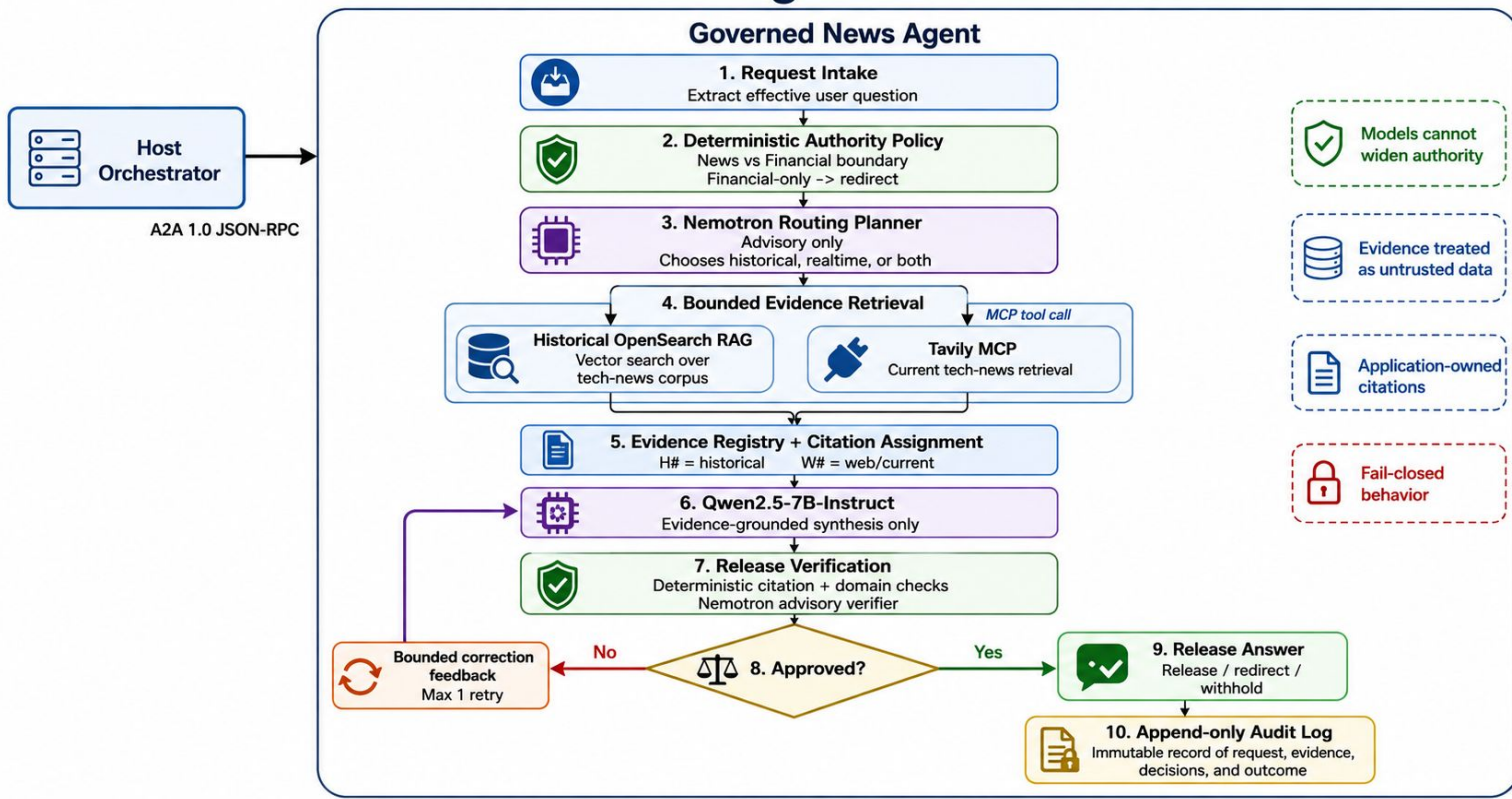
- LLM Doesn't Own the Answer
 - Retrieve Evidence
 - Provide Assessment
- Nemotron Orch. Owns:
 - Advisory Role
 - Deterministic Check
 - "Second Opinion" (opt.)
 - Policy Verifier
 - Can End Workflow



News Agent Architecture

Image Attribution
Gemini Generated

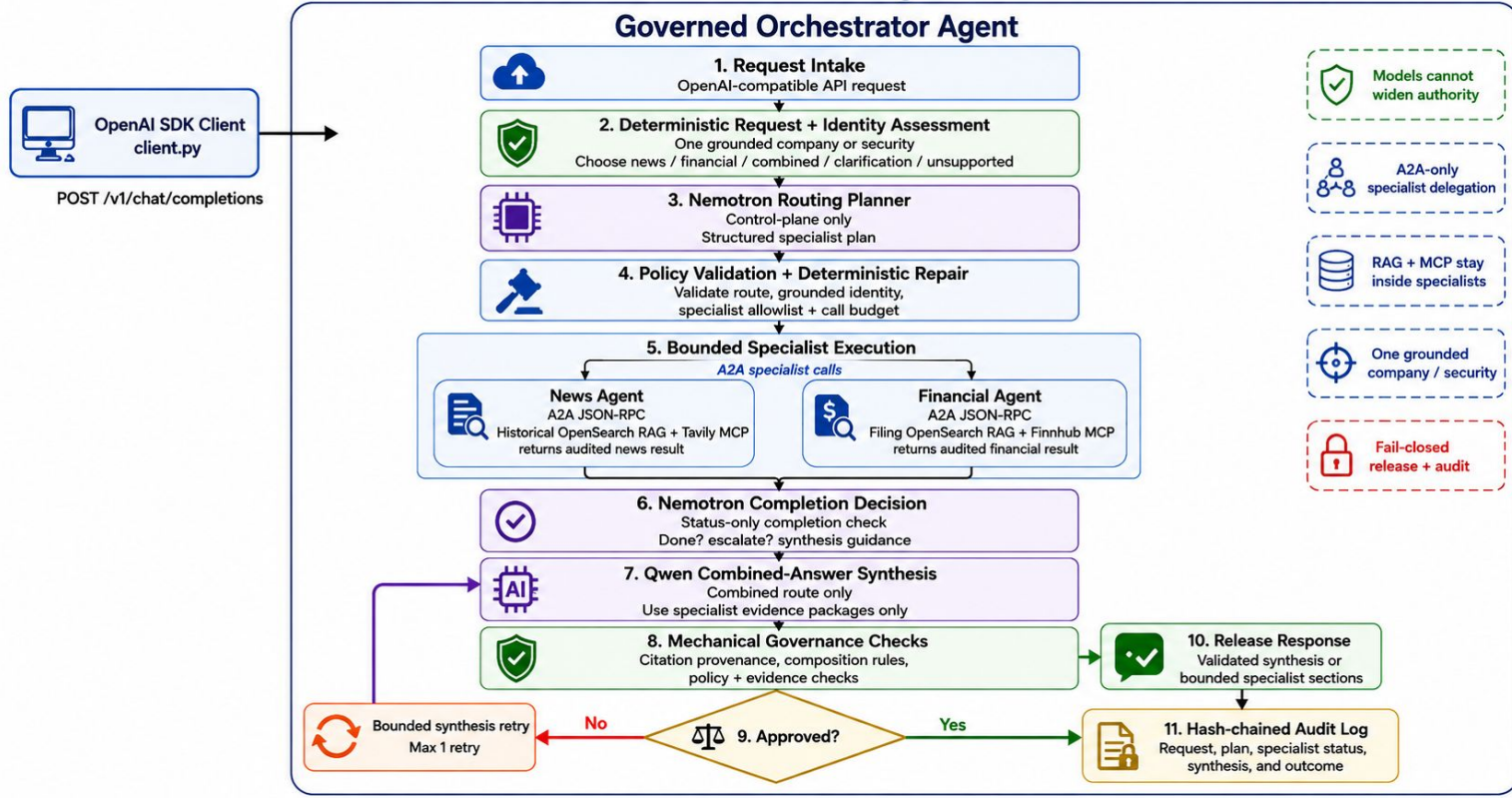
Governed News Agent Architecture



Orchestrator Architecture

Image Attribution
Gemini Generated

Governed Orchestrator Agent Architecture

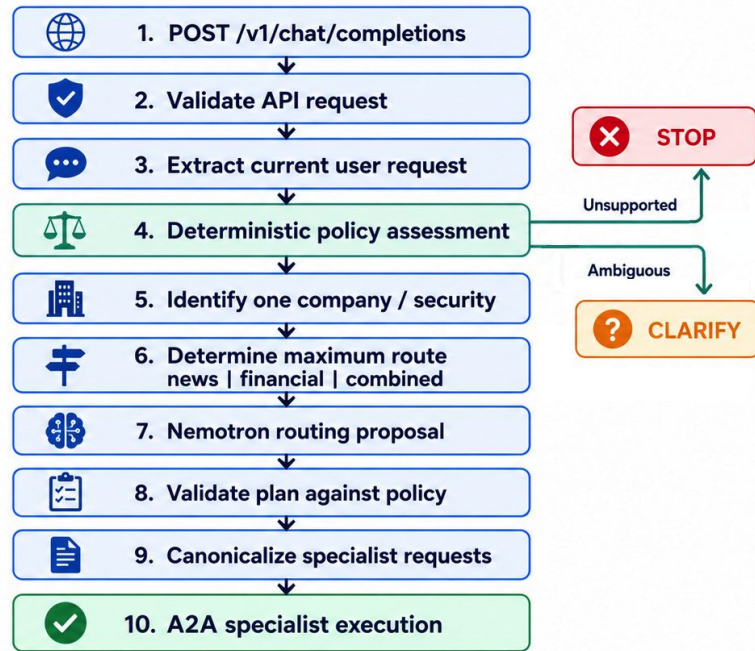


Who Plays a Role in the Question?

Intent Based Routing!

- Uses Nemotron Orch
- News:
 - Announcements
 - News, Products, Strategy
 - Partnerships, Acquisition
 - And so on...
- Financial:
 - Stock, tick, earnings, etc
 - Revenue, SEC, Financials
 - And so on...

Input Validation and Routing Flow



Routing Decision Matrix: Examples

User Asks...	Route	Specialist Evidence
What is NVIDIA doing in AI?	News	Historical OpenSearch + Real-Time Data (as needed)
What is NVDA's current stock price?	Financial	Real-time Stock Quote via MCP
How did NVIDIA perform last quarter?	Financial	SEC RAG via OpenSearch + Real-time Earnings (as appropriate)
What is NVIDIA doing in AI and how are its financials?	Both	News and Financials Independently
What is the weather?	Unsupported	No specialist called

Chain Of **DATA** Custody

News Agent

- |
- +--> [H1] Historical News 1
- +--> [W1] Current News 1
- +--> [W2] Current News 2

Financial Agent

- |
- +--> [F1] SEC filing Q1
- +--> [F2] SEC filing Q2
- +--> [M1] Finnhub Market Data 1
- +--> [M2] Finnhub Market Data 2

You: Can you tell me what NVIDIA is doing in
Response time: 200.91 seconds

Orchestrator:

NVIDIA is expanding its role in the artificial intelligence ecosystem by quietly releasing the Llama-3.1-Nemotron-70B, along with the introduction of the open-source AI software [H2]. The company is also in partnership with Carnegie Mellon University and the University of Texas at Austin for AI infrastructure efforts [H3].

These developments have contributed to sustained growth in the semiconductor sector, expected to continue fueling growth, with one analyst projecting a demand and massive revenue backlogs at major hyperscalers to less than \$120 billion to manage

As of August 26, 2026, NVIDIA's stock (NVDA) has shown a strong upward trajectory, despite short-term volatility [M1]. This growth is directly influencing investor expectations for future performance.

***Specialist audit IDs:** `financial=083a6c78`

***Routing trace:** `route=combined | company=NVIDIA`

```
237     "output_length": 1,
238     "output_sha256": "a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6q7r8s9t0u1v2w3x4y5z6",
239     "released": true,
240     "requested": "2026-08-27T02:58:32.897057+00:00",
241     "source_citations": [
242         "[H2]",
243         "[H3]",
244         "[H4]",
245         "[H5]",
246         "[M1]",
247         "[W1]",
248         "[W2]",
249         "[W3]",
250         "https://finbold.com/ai-predicts-nvidia-stock-price",
251         "https://finnhub.io/",
252         "https://seekingalpha.com/news/4633594-nvidia-scale",
253         "https://venturebeat.com/ai/nvidia-just-dropped-a-n",
254         "https://venturebeat.com/ai/nvidia-showcases-tech-i",
255         "https://www.crn.com/news/networking/2024/nvidia-hi",
256         "https://www.fool.com/investing/2026/08/21/predicti",
257     ],
258 },
259 "timestamp": "2026-08-27T02:58:32.897057+00:00"
260 }
```

Image Attribution

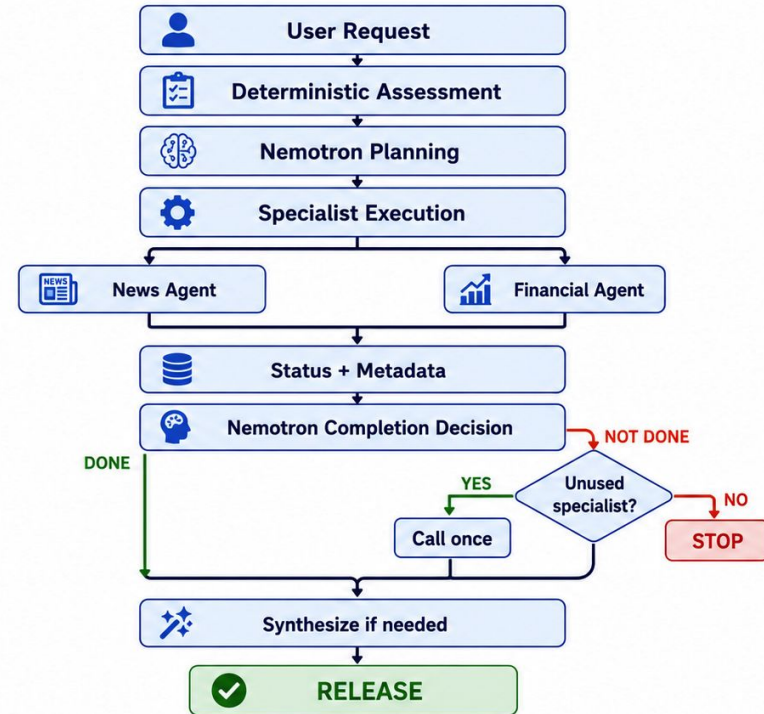
Princeton University Researcher - [Tool helps clear biases from computer vision](#)

Molly Sharlach on October 1, 2020

Answer: Synthesize a Result

- When Are We Done?
 - Policy/Contract is Met
 - Specialists Executed
 - Within "Calling" Budget
- Synthesized = Combined
 - All Specialists Data Synthetically "Combined"
 - Think Of This As Memory Compression (Same Problem)

Completion and Release Flow



Demo

<https://youtu.be/OVjpcicTMLQ>

Chain Of Data Custody
+
Policy Enforcement
=
AI Governance

And By The Way...
Mixture of Experts (MoE)
Are Just Vertical APIs

Resources

Resources

SCAN THE QR CODE FOR ALL RESOURCES!!

GitHub: github.com/davidvonthenen/2026-api-world

Check Out [NetApp Instacluster](#)

- OpenSearch: RAG using Vector and BM25
- Kafka: Data Streaming and Event-Driven Arch
- Cassandra: Hybrid RAG (Graph-like + Vector)
- ClickHouse: AI Governance (Audit, Observability)



Thank You!



David vonThenen
Senior AI/ML Engineer



[@davidvonthenen](https://twitter.com/davidvonthenen)

